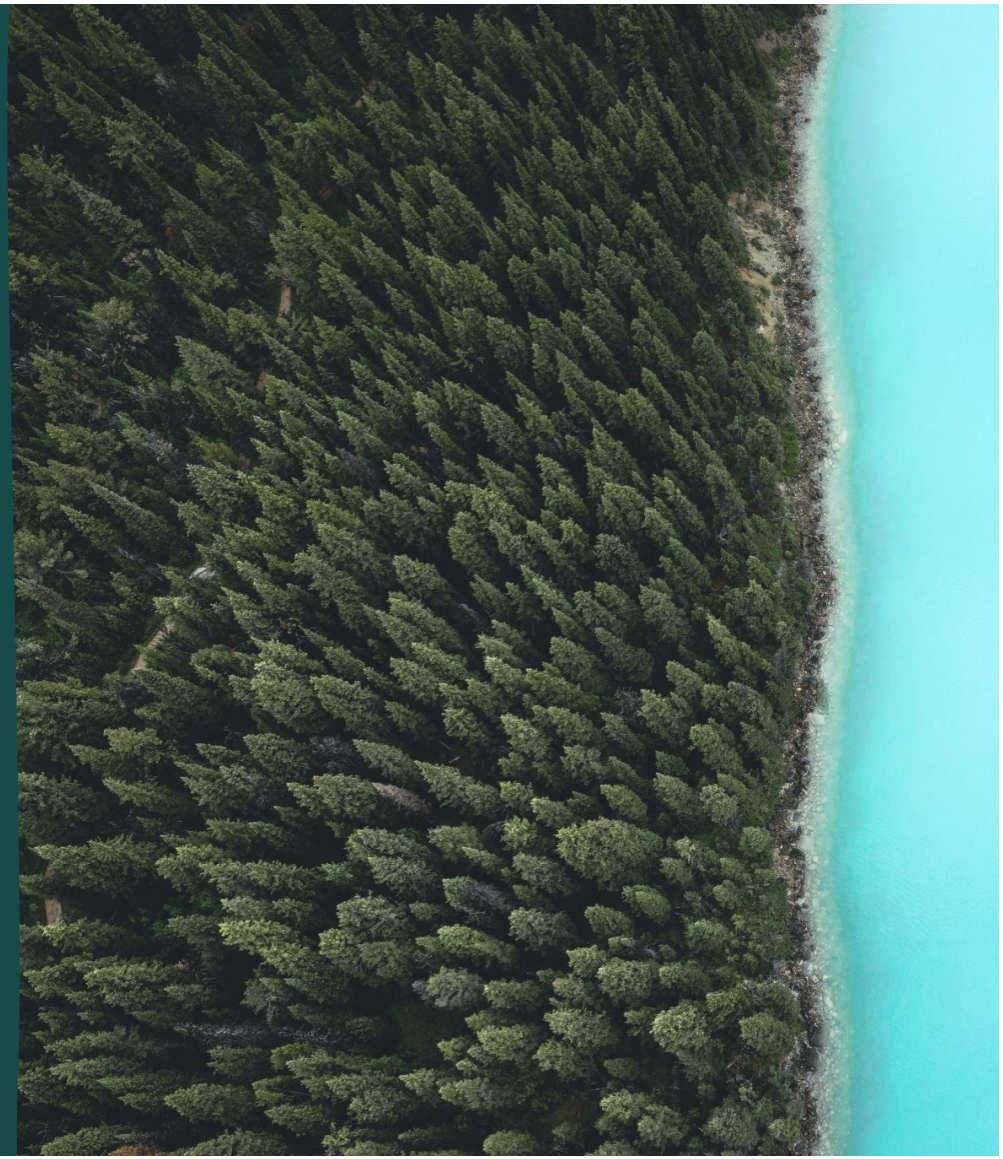
HEALTH DISPARITIES & TECHNOLOGY

COLLEGE 117

April 20, 2026

Dr. Martinez

Dr. Martschenko



Overview

- Last week:
  - Social Determinants Shape Health
  - Social Determinants impact the level of health that a community or group of people can attain
  - Social determinants create health injustice
- This week:
  - How scientific research and the distribution of resources for research impact health research and health disparities

Technology & Health Research

- Inequity in health research impacts healthcare
- Inequity in health research impacts the development of technology
  - Which, in turn, impacts healthcare → and can further impact the technology

Daniels

- A health inequality is unjust when it derives from an unjust distribution of the socially controllable factors affecting population health and its distribution

Knowledge infrastructure

- “Robust networks of people, artifacts, and institutions that generate, share, and maintain specific knowledge about the human and natural worlds.” – Paul Edwards (STS)
  - Relationships/networks
  - Includes multiple systems
  - Examples: Weather Forecast infrastructure; CDC



What are elements of scientific research infrastructure?

Infrastructure for Biomedical Research



Funding



Institutions

Universities

Hospitals



People

Scientists/ researchers

Elements of Scientific Research Infrastructure

Funding: Government, Private Foundations, Companies

Research Institutions: Universities, Industry

Oversight: Government, Institutional Review Boards

Education/Training: Colleges/Universities

Public Engagement: Media

Dissemination: Publications, Databases

Technology Production

COVID Infection Rates

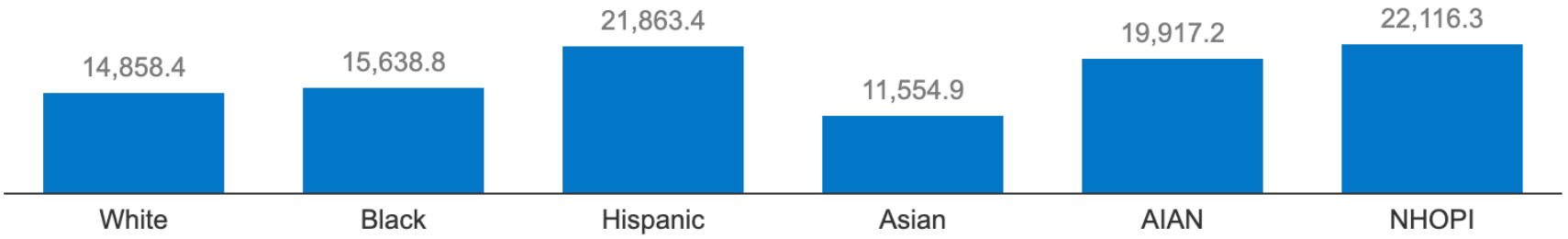
Figure 1

Cumulative COVID-19 Age-Adjusted Infection Rates by Race/Ethnicity, 2020-2022

Rates per 100,000 population

Click on the buttons below to see data for the different metrics:

Cases Deaths



NOTE: Persons of Hispanic origin may be of any race but are categorized as Hispanic for this analysis; other groups are non-Hispanic. AIAN refers to American Indian or Alaska Native. NHOPI refers to Native Hawaiian or Other Pacific Islander. Case data as of August 1, 2022. Age-adjusted rates standardized to 2019 U.S. Census Bureau population estimates.

SOURCE: KFF analysis of Centers for Disease Control and Prevention, COVID-19 Response. COVID-19 Case Surveillance Restricted Data Access, Summary, and Limitations, released on August 4, 2022. The CDC does not take responsibility for the scientific validity or accuracy of methodology, results, statistical analyses, or conclusions presented. • PNG

KFF

Covid Death Rates

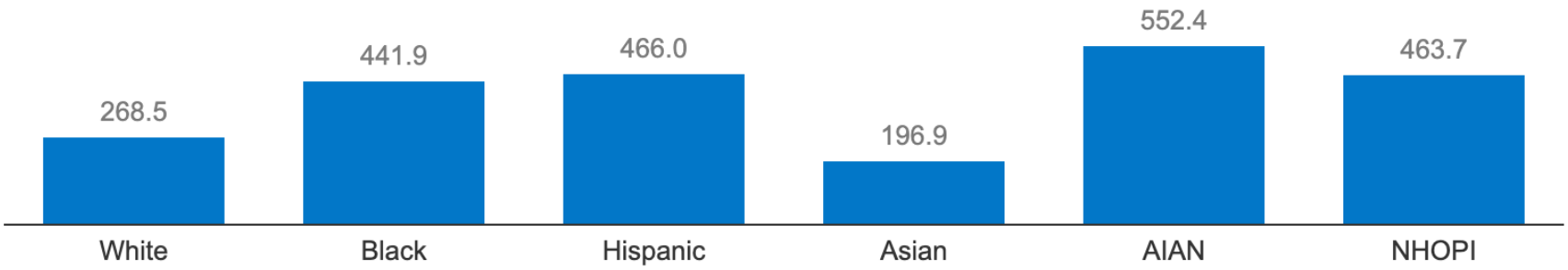
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SOURCE: KFF Analysis of National Center for Health Statistics. Provisional COVID-19 Deaths by HHS Region, Race, and Age. Date accessed August 4, 2022. Available from https://data.cdc.gov/d/tpcp-uiv5. • PNG

KFF

U.S. NEWS > RACE IN AMERICA

Filipino nurses died at disproportionate rate during pandemic

Filipino nurses were some of health care's heroes during the pandemic, but their health and heritage is often looked over.

Filipino nurses make up 4.5% of the nursing population but account for 25% of COVID deaths.

COVID-19 AND MEATPACKING

COVID-19 cases, deaths in meatpacking industry were much higher than previously known, congressional investigation shows

A House of Representatives subcommittee reviewing the coronavirus pandemic found plants owned by five major meatpacking companies accounted for at least 59,000 COVID-19 cases and 269 deaths — much higher than any previous tally.

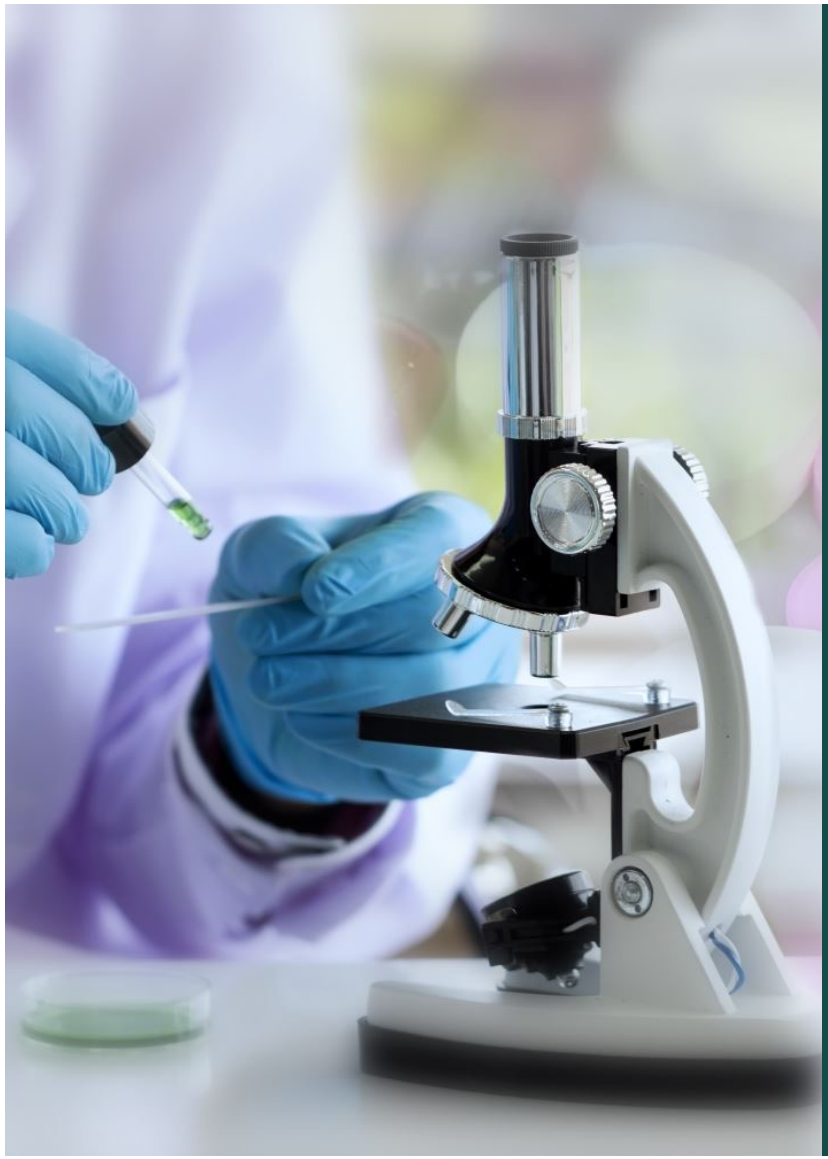
What is a potential value/benefit of a country funding scientific research?

Some Outcomes from Scientific Research Infrastructures

- Direct: Scientific Knowledge
- Societal benefits
  - Social innovation
  - Positive community impact – economic stability
  - Responses to challenges and crises
  - Positive impacts on the whole scientific ecosystem
- Risks/harms
  - Unequal access/benefits
  - Environmental harms

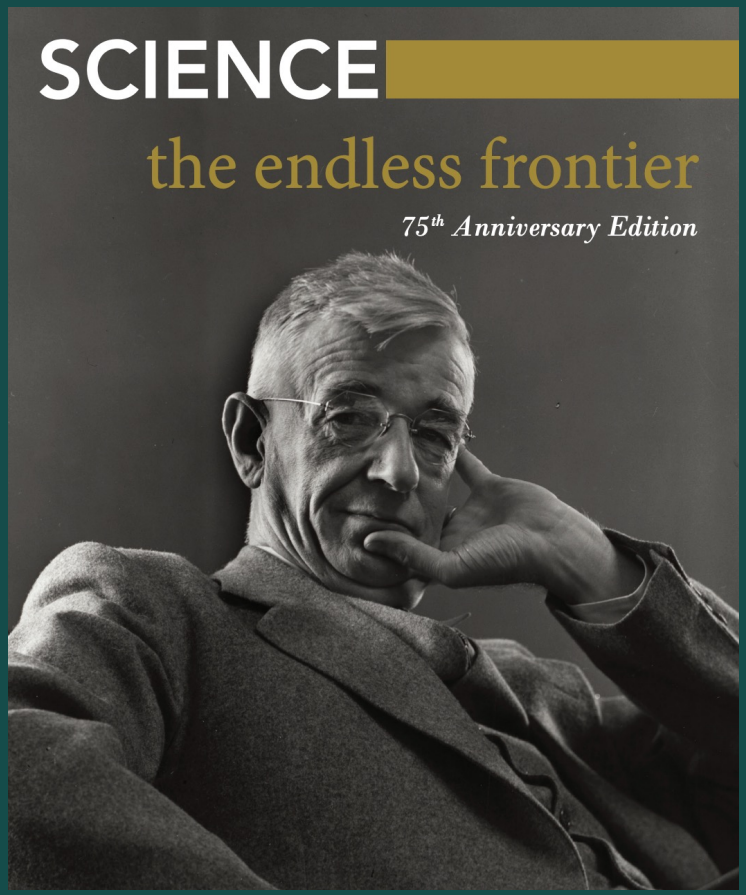


Who/where steers what scientific research is conducted?
Who/where steers what scientific questions are answered and how they are answered?



Scientific research & healthcare

- Funding for biomedical researcher concentrated in the upper-income countries
  - Impacts priorities for health research
  - Understanding different health conditions
  - Researchers & research capacity

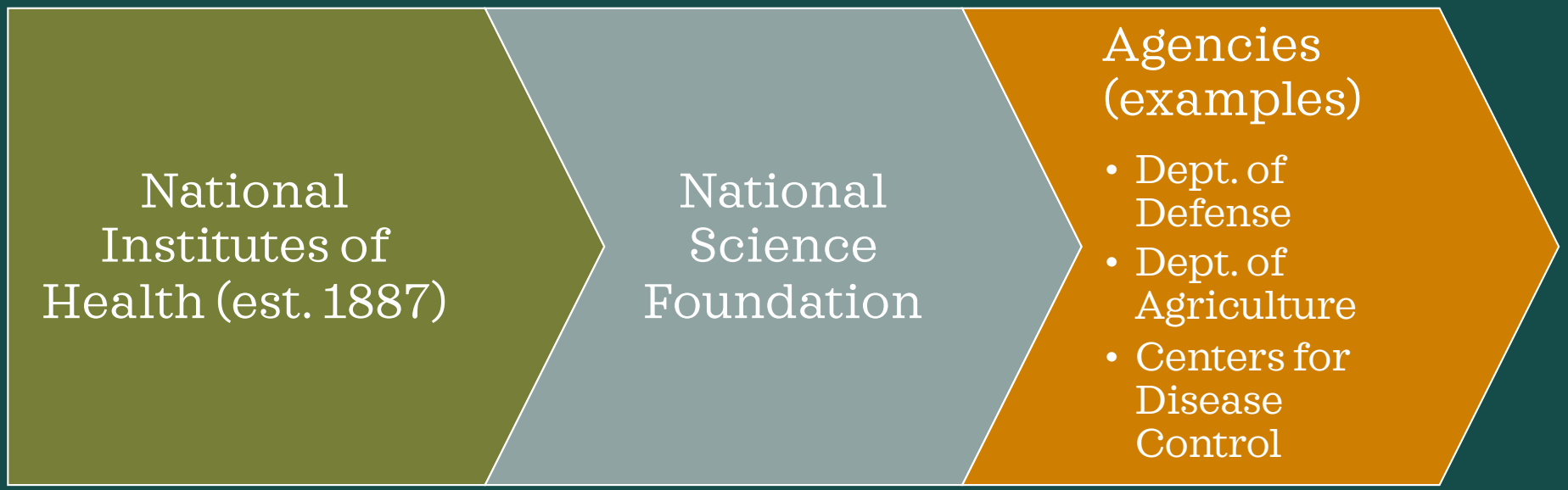


National Science Foundation

ESTABLISHED 1950

FUNDS BIOMEDICAL RESEARCH AS FOUNDATION FOR U.S. SECURITY AND PROSPERITY

US Federal Science Funding





Rural State Impact Graphic

When researchers and organizations in a state are awarded funding from the National Institutes of Health (NIH), the impact of that funding reaches far beyond its original recipients.

United for Medical Researc

NIH or NSF funding helped* with the discovery or development of....



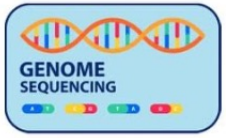
The internet



MRI



Kidney matching



The human genome project



Medications for autoimmune disease



Better survival for childhood ALL (a type of leukemia)

*these are select examples, my list could be much longer.

NIH FUNDING

CREATES JOBS, MEDICINES, & ECONOMIC GROWTH

THE NIH PREVENTS THE U.S. FROM FALLING BEHIND IN BIOMEDICAL RESEARCH

NIH GRANT $ GROW THE ECONOMY

$1 from NIH = $2.46 in economic activity



99.4% OF MEDICINES FDA APPROVED WERE FUNDED BY NIH



NOT ALL GRANT OVERHEAD IS THE SAME

Other grants allow costs the NIH restricts



NIH DIRECTLY SUPPORTS JOBS



These are the faces of people with NIH funded jobs

THEY DEVELOP CURES FOR CANCER, SAFER VACCINES & TRANSFORMATIVE THERAPIES FOR AUTOIMMUNE DISEASES

NIH FUNDS DISCOVERIES IN INDUSTRY



NIH funds ~43% of costs to bring medicines to patients

Clearly et al. JAMA Health Forum. 2023 * DiMasi et al. J Health Econ. 2016 * https://www.unitedformedicalresearch.org/annual-economic-report/

Disparities in how funding is allocated

- Types of research questions funded
- Institutional level: which institutions get funding
- Underrepresentation among scientists

Study shows Black researchers about 55% less likely to receive NIH funding than white researchers

by Navya Gupta
February 5, 2021

RESEARCH ARTICLE | SCIENTIFIC COMMUNITY

Topic choice contributes to the lower rate of NIH awards to African-American/black scientists

TRAVIS A. HOPPE, AVIVA LITOVITZ, KRISTINE A. WILLIS, REBECCA A. MESEROLL, MATTHEW J. PERKINS, B. IAN HUTCHINS, ALISON F. DAVIS, MICHAEL S. LAUER, HANNAH A. VALANTINE, [...], AND GEORGE M. SANTANGELO +1 authors Authors Info & Affiliations

SCIENCE ADVANCES • 9 Oct 2019 • Vol 5, Issue 10 • DOI: 10.1126/sciadv.aaw7238

Black scientists are more likely to work in research areas with less NIH funding

by ***Andrea Widener***

October 12, 2019 | A version of this story appeared in **Volume 97, Issue 40**

RESULTS

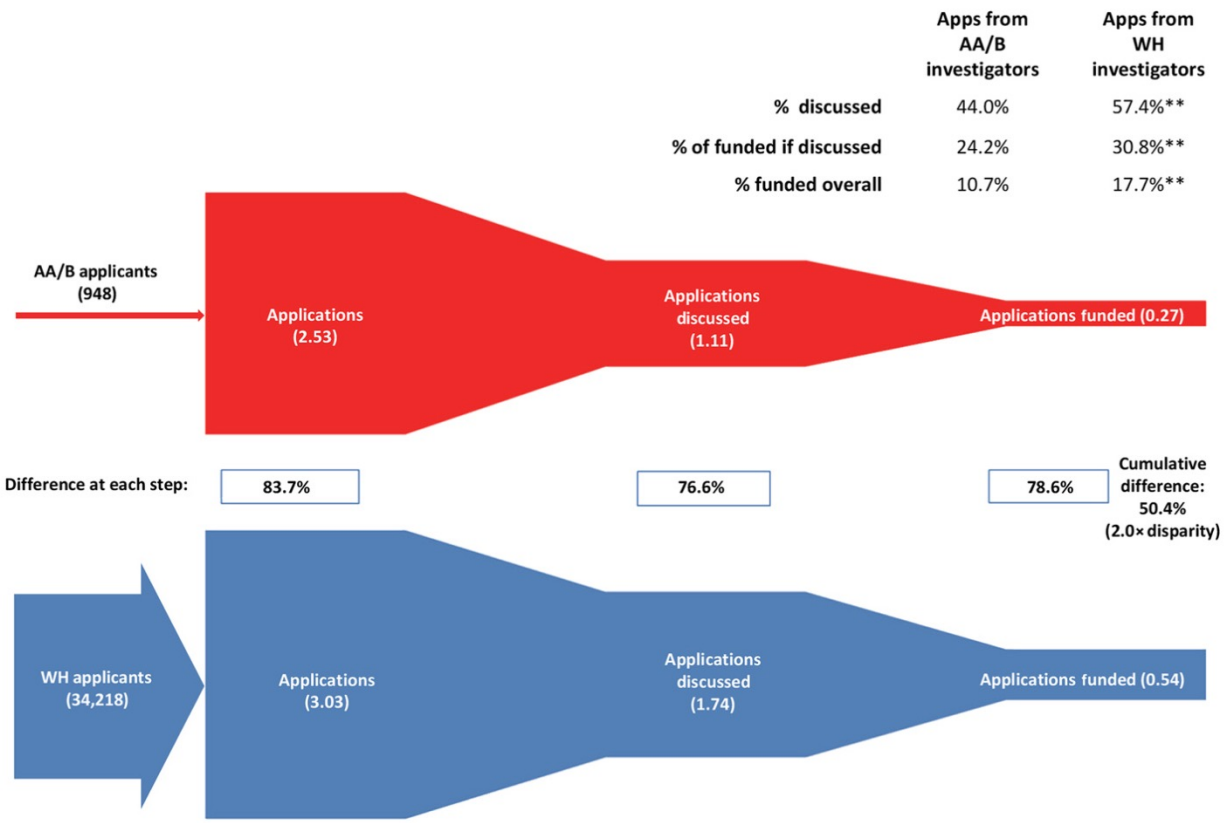


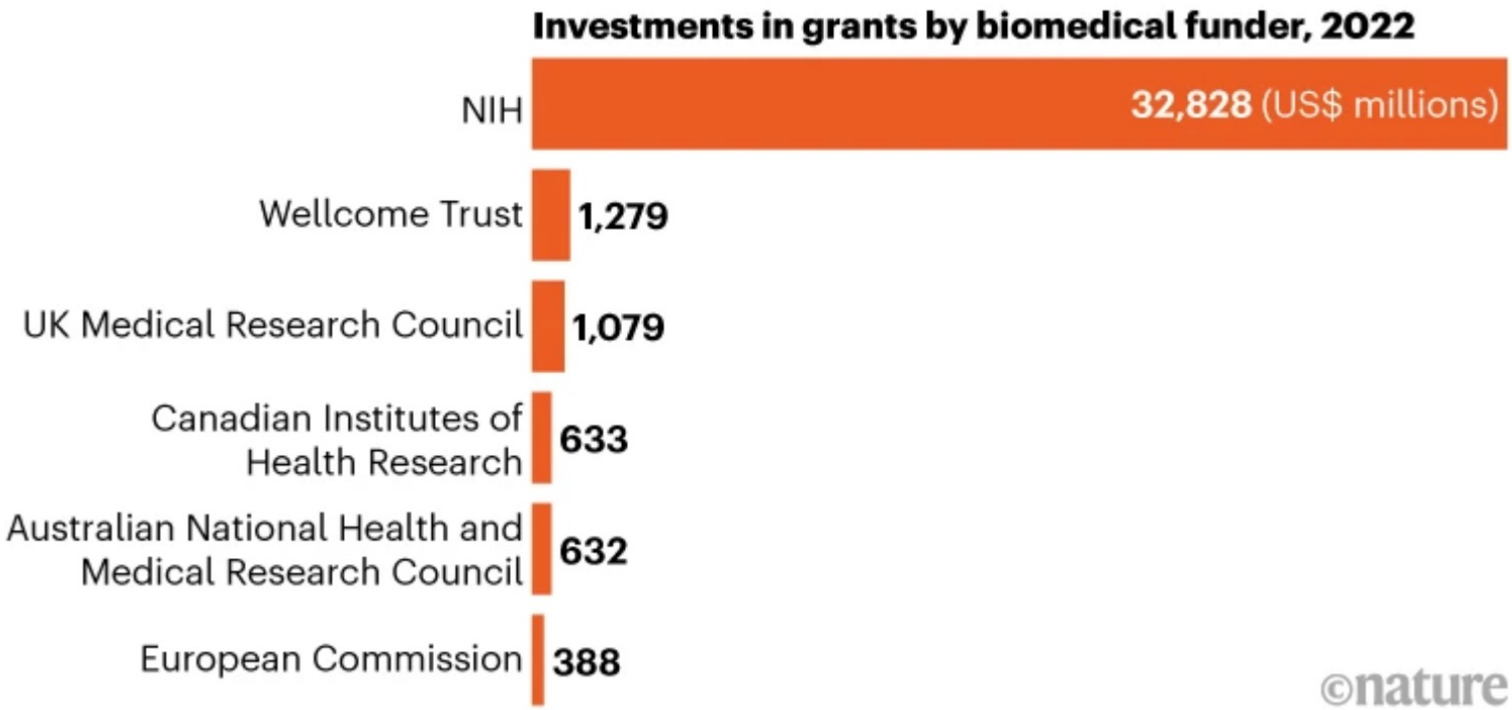
Fig. 1 Funding gap between AA/B and WH scientists at each stage of the R01 application and review process.

Arrows on the left indicate the number of AA/B and WH R01 applicants in FY 2011–2015. The total number of applicants with a reported race/ethnicity is 45,998. Rocket charts depict the number of applications that were submitted, discussed, and funded per applicant. Comparative rates of discussion, funding of discussed applications, and overall funding rates are presented on the top right (**$P < 0.01$).

What is a potential value/benefit of ensuring that scientific research is being conducted globally?

WORLD LEADER

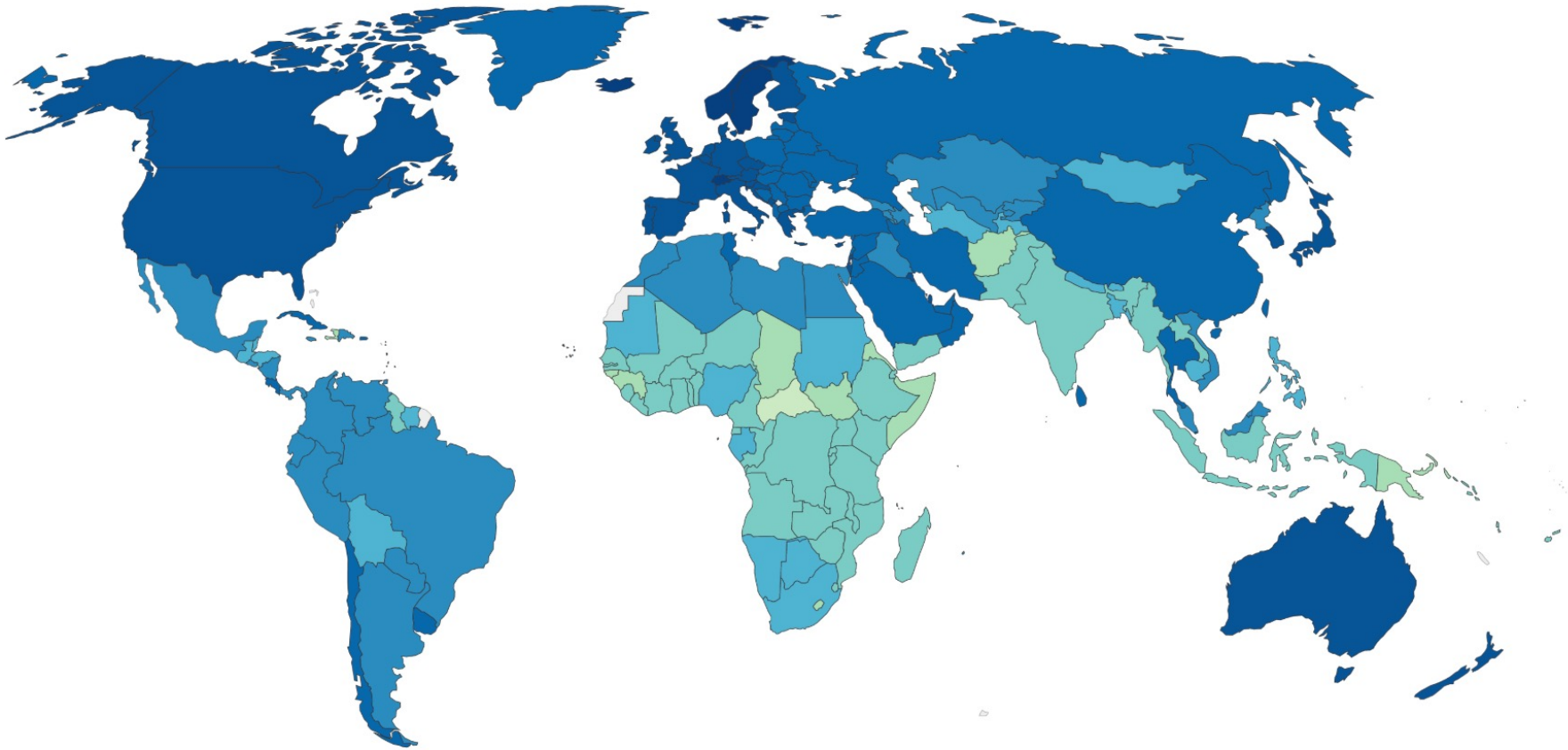
The US National Institutes of Health (NIH) has a budget for biomedical-research grants that towers over the world's next-biggest funders of health research.



Source: World RePORT

Healthcare Access and Quality Index, 2015

The Healthcare Access and Quality (HAQ) Index is measured on a scale from 0 (worst) to 100 (best) based on death rates from 32 causes of death that could be avoided by timely and effective medical care (also known as 'amenable mortality').



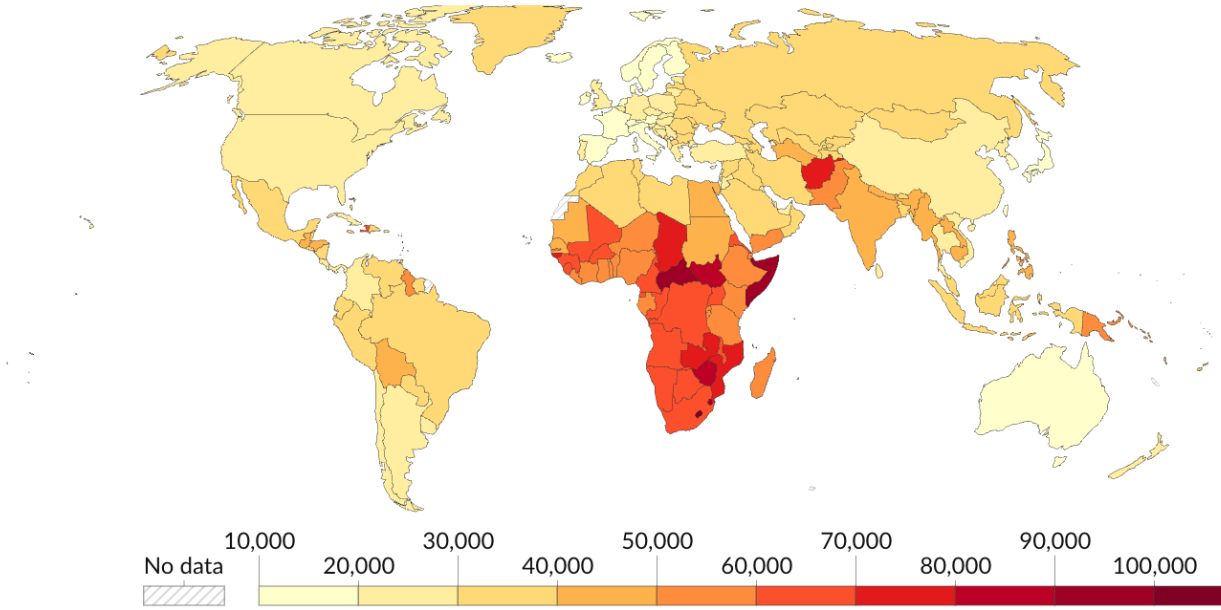
Burden of disease, 2021

Our World in Data

Disability-Adjusted Life Years (DALYs) per 100,000 individuals from all causes. DALYs measure the total burden of disease – both from years of life lost due to premature death and years lived with a disability. One DALY equals one lost year of healthy life.

Table | Map | Chart

World



1990 — 2021

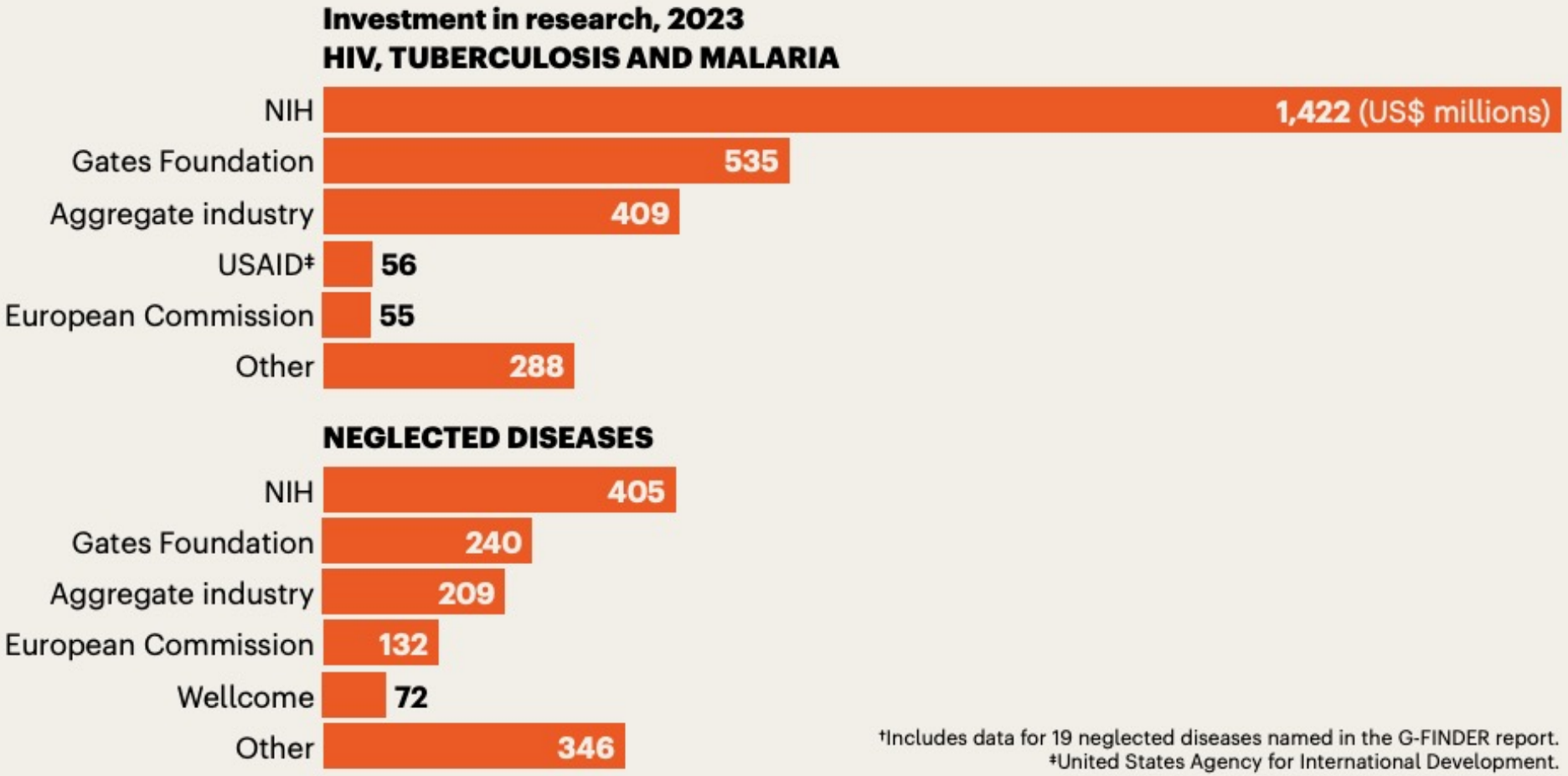
Data source: IHME, Global Burden of Disease (2024) – Learn more about this data

Note: To allow for comparisons between countries and over time, this metric is age-standardized.

OurWorldinData.org/burden-of-disease | CC BY

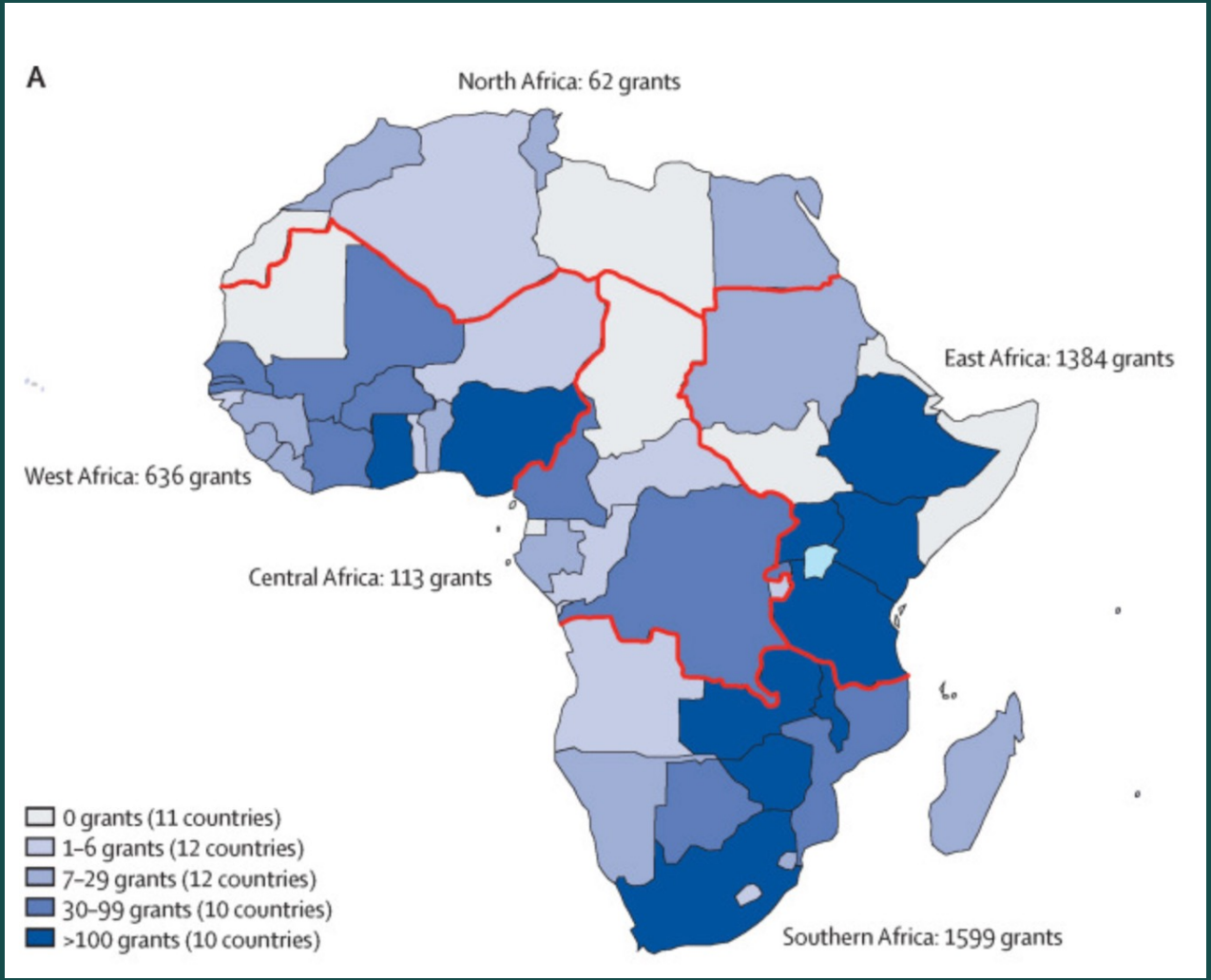
TACKLING GLOBAL DISEASES

The NIH funds most of the world's science on diseases that are major killers, such as HIV, tuberculosis and malaria, and a vast portion of the research on neglected diseases such as snakebite, scabies and leprosy†.



†Includes data for 19 neglected diseases named in the G-FINDER report.
‡United States Agency for International Development.

SOURCE: G-FINDER PROJECT



Biomedical Research Funding in African Countries (2017)

World RePORT: a database for mapping biomedical research funding
Adam, Taghreed et al.
The Lancet Global Health, Volume 8, Issue 1.

Research funding Trends in Research On Ebola and Sickle Cell Disease, 2012-7

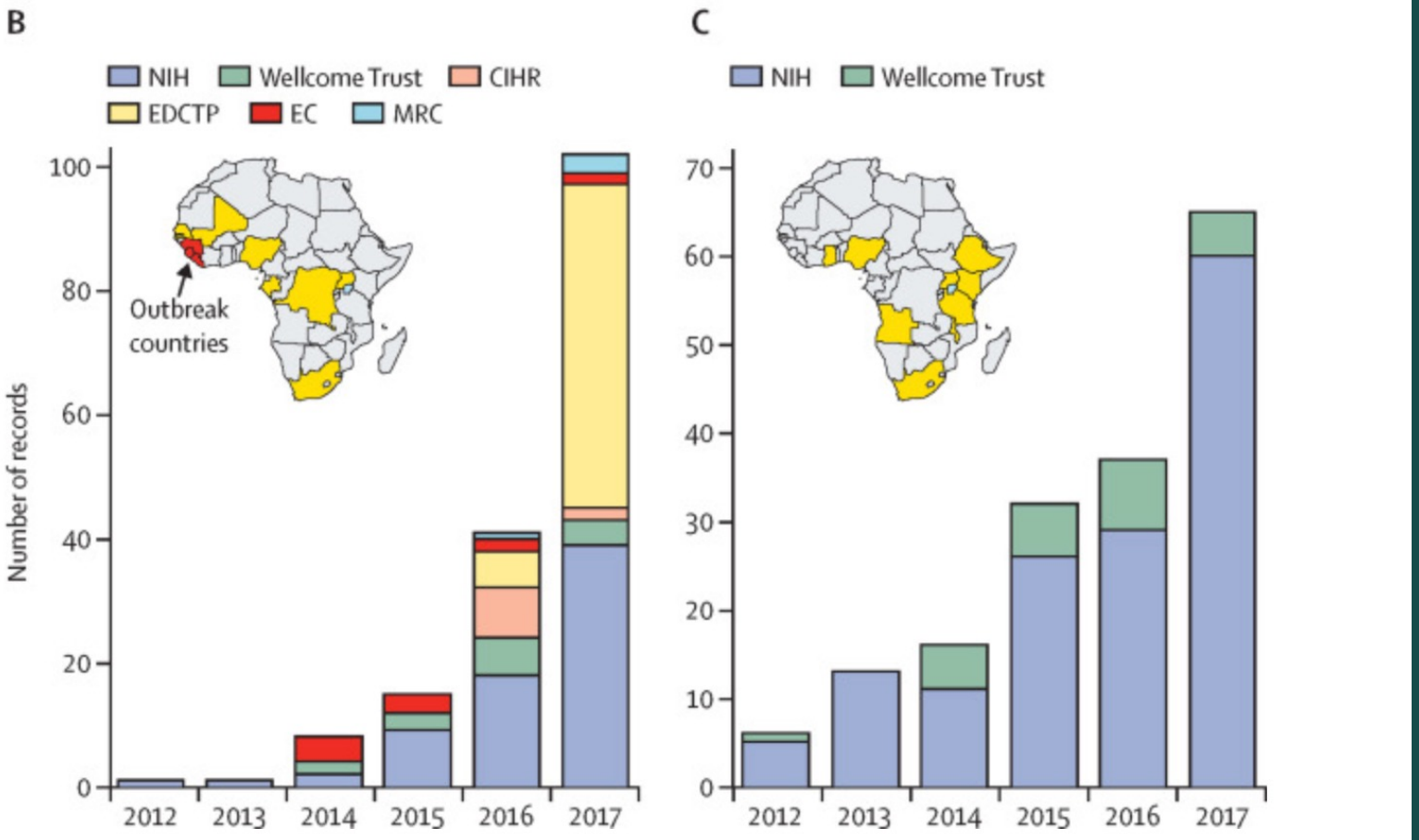
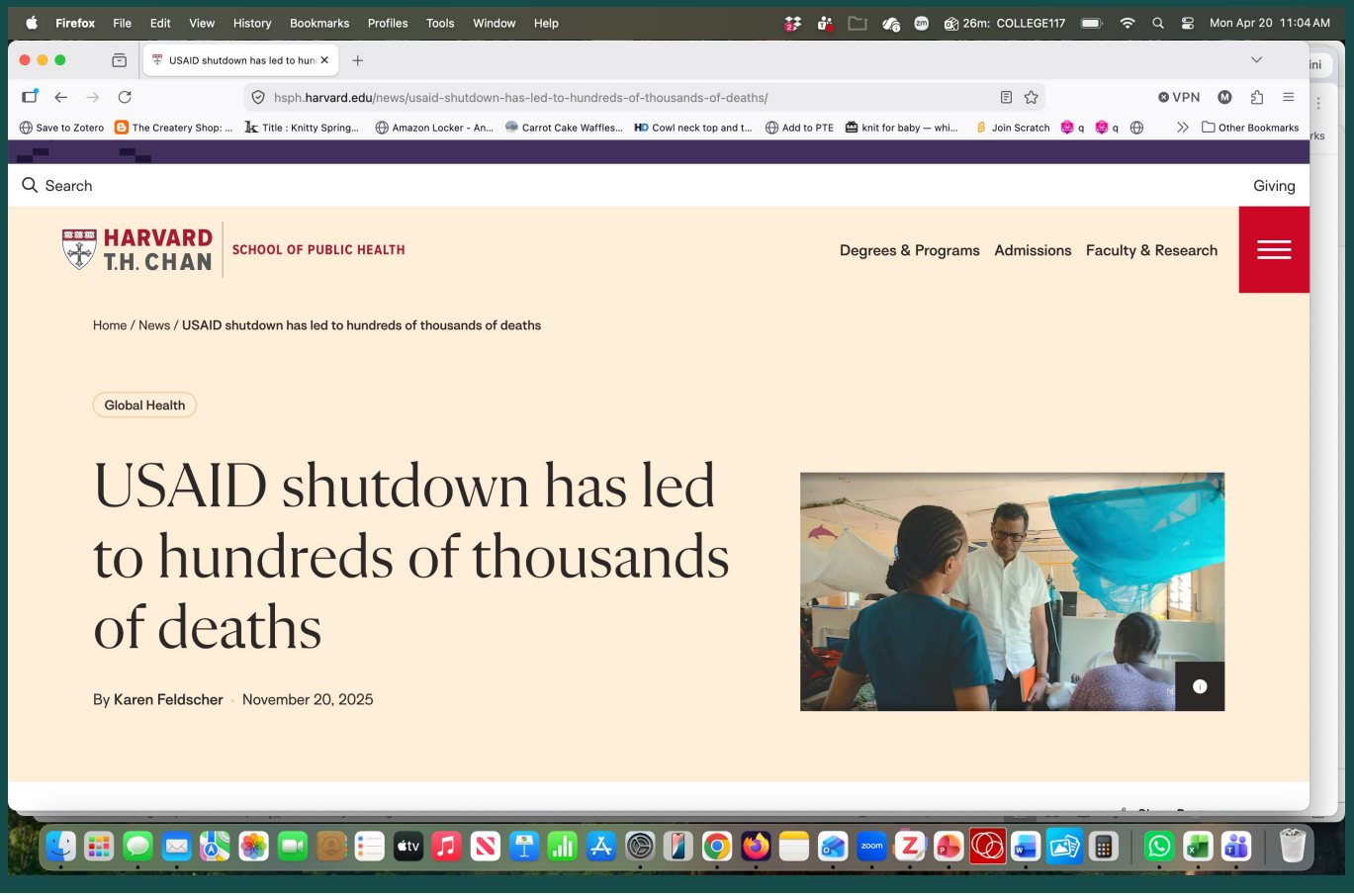


Figure Grants in Africa by region, 2017 (A), and trends in research on Ebola (B) and sickle cell disease (C), 2012–17

Consequences of USAID funding cuts



How is scientific research relevant to health disparities?

Global Health Research & Health disparities

- Global health research should generate knowledge to reduce health disparities (Pratt)
- Lack of national public health and biomedical data from research in Africa poses challenges in addressing the health burden (Mulder, et al, 2021)

- How do research and biotechnology infrastructure impact justice concerns in health care?

Cahill: Biotech & Justice

- Bioethics as institution
  - US & Europe –biomedical research & funding
    - Bioethics initially developed & funded to prioritize US/European values

Bioethics: 4 Principles (Belmont Report)

- Autonomy – historically, has been prioritized in U.S. bioethics
- Beneficence & Non-maleficence
- Justice

Cahill: Biotech & Justice

- Need for more attention to:
  - Justice
  - International cooperation & governance
- Example: HIV drugs in South Africa